TG ECET (WP)-2026

SYLLABUS FOR COMPUTER SCIENCE & ENGINEERING

MATHEMATICS (10 Marks)

S.No	Syllabus
1	**UNIT I: MATRICES AND DETERMINANTS** Definition of Matrix-order of a matrix-Type of matrices - Algebra of matrices- sum, difference, scalar multiplication and product of 2x2 Matrix - Equality of two matrices-Transpose of a Matrix – Determinant of a square Matrix of order 2×2.
2	**UNIT II: CO-ORDINATE GEOMETRY** Straight lines: Slope of straight line - Various forms of straight line: Point-Slope form, Two- Point form, Intercept form, and General equation of a Straight-line Conditions for Parallelism and Perpendicularity of the Straight lines – Equation of a straight line parallel and perpendicular to the given straight line -Distance between two Parallel lines.
3	**UNIT III: TRIGONOMETRY** Trigonometric Ratios of Compound angles (without proof) - Trigonometric Ratios of Multiple and Sub- multiple angles (2A, 3A, A/2).

COMPUTER SCIENCE & ENGINEERING (90 Marks)

S.No	Syllabus
1	**Unit-I: DIGITAL ELECTRONICS** Number systems–Number conversions– Codes – Logic gates: AND, OR, NOT, NOR, NAND and XOR – Boolean algebra – Boolean expressions – De-Morgan's theorems – SOP and POS forms –K-Map (up to 4 variables) –Digital logic families – TTL, CMOS and ECL–Characteristics of Digital ICs –Combinational Circuits –Adders – Multiplexers and De-multiplexers –Encoders and Decoders –Comparators –Sequential logic circuits –Latches – Flip-flops – Edge and level triggering –Registers –Counters –Memories – RAM, ROM, Applications of Flash ROM.
2	**Unit-II: COMPUTER ARCHITECTURE** Functional blocks of Digital computer – Stored program concept – Fixed point, Floating point number representations – Complements –Instruction formats – Addressing modes–Memory hierarchy –Virtual memory, Associative memory – Cache memory – I/O Organization – Modes of data transfer – Programmed I/O, Interrupt initiated I/O, and DMA – Bus system – Parallel processing –Pipeline processing: Arithmetic & instruction pipeline – Vector processing – Flynn's classification – Comparison between RISC and CISC processors.

3	**Unit-III: C PROGRAMMING AND DATA STRUCTURES** Algorithms – Flowcharts – C Tokens – Data types – Operators – Expressions – Precedence and Associativity of operators –Type conversions – Preprocessor directive statements– Decision making statements –Looping statements – break and continue – 1D and 2D Arrays –Strings – String handling functions – Scope and lifetime of variables– Functions– Parameter passing– Storage classes–Recursion – Structures– Unions – Files: File opening modes, Open/Closing files. **Data Structures:** Pointers – Address of(&) and Dereferencing(*) operators–Pointer Arithmetic– Arrays and pointers– Structures and pointers – Dynamic Memory allocation– Data structures classification– Abstract Data Types– Asymptotic Notations: Big-Oh, Omega, Theta– Stacks and Queues –Linked Lists: Single Linked List, Double Linked List, Circular Linked List– Trees–Binary trees– Tree traversal techniques. Sorting Techniques: Bubble, Selection, Insertion, Quick and Merge sort– Searching Techniques: Linear and Binary search.
4	**Unit-IV: OBJECT ORIENTED PROGRAMMING THROUGH C++** OOPs concepts – Keywords of C++ –Classes and objects– Array of objects– Passing and returning objects– Pointer to objects– References– this pointer– I/O manipulators – File and I/O functions –Constructors and destructors –Function overloading, Constructor overloading and Operator overloading– Inheritance types: Single, Multiple, Multilevel, Hierarchical, Hybrid and Multipath–Virtual functions – friend functions –inline functions – Templates: function and class templates.
5	**Unit-V: RELATIONAL DATABASE MANAGEMENT SYSTEMS** Database System Concepts – Data Abstraction – Data Independence– Data Models– Entity-Relationship (ER) Model – Structure of Relational database – Types of Keys– Functional dependencies – Normal Forms: 1^{st}, 2^{nd}, 3^{rd} and BCNF – Transactions – ACID Properties – SQL – data types, operators, constraints – Database Languages: DDL, DML – SQL functions: numeric, aggregate, scalar, date and string functions – JOIN statements – views, sequences, synonyms and indexes – PL/SQL: data types, control statements, functions, procedures, recursion, exceptions, cursor management, and triggers.
6	**Unit-VI: COMPUTER HARDWARE & NETWORKING** BIOS – Components of Motherboard – SMPS –Processors – Memories –Mass Storage devices –Input devices– Output devices. **Networking:** Classification of networks: LAN, MAN, WAN –Network topologies: Bus, Ring, Star, Mesh, and Hybrid –OSI reference model – TCP/IP reference model – LAN components: Coaxial, Twisted-pair, Optical fiber cables and connectors – Ethernet – LAN devices: Repeaters, Hubs, Bridges, Switches, Gateways, Network Interface Cards(NICs), Routers, Modems – Protocols: HTTP, HTTPS, FTP, SMTP, Telnet –TCP/IP addressing scheme – IP Address classes – IP Subnetting.

7	**Unit-VII: OPERATING SYSTEMS** Operating System Concepts – Goals– Services– Types– System calls – Process Management: PCB, Process states, Threads – CPU Scheduling criteria –CPU Scheduling Algorithms: FCFS, SJF, Round Robin, Priority, Multilevel Scheduling, Multilevel Feedback Scheduling – Inter Process Communication –Process Synchronization – Semaphores – Monitors – Deadlocks: Necessary conditions, Prevention, Avoidance, Detection and Recovery – Memory Management – Overlays, Swapping, Fragmentation, Paging, Segmentation - Virtual Memory– Demand Paging – Page Replacement Algorithms: FIFO, LRU, Optimal – Thrashing – Disk Scheduling – Disk Scheduling Algorithms: FIFO, SSJF, SCAN, C-SCAN – File management: File operations, Access methods, Directory Structure.
8	**Unit-VIII: JAVA PROGRAMMING** Java – features, tokens, data types, variables, operators, arrays, selection and iteration statements – Classes and objects – Constructors – Method overloading –Static and final usage – String class and String class methods– Inheritance types – this, super keywords – Method overriding – Interfaces –Packages – Access specifiers: public, private, protected, default – Applets – AWT – AWT Controls: Label, Button, Checkbox, CheckboxGroup, TextField, TextArea, Choice, List, Scrollbar – Event Sources, Event Classes, Event Listener Interfaces – Event handling – Exception handling – Multithreading – JDBC – Servlets.
9	**Unit-IX: PYTHON PROGRAMMING** Python – features, variables, data types, indentation, decision making statements, looping statements – Data Structures: Lists, Tuples, Sets and Dictionaries –Classes and objects – Constructors – Modules – Packages: math, datetime package– Exception handling – Multithreading– GUI using Tkinter package – Geometry managers: Pack, Grid, Place – Widgets: Label, Entry, Text, Button, Checkbutton, Radiobutton, Listbox, Canvas, Scale, Scrollbar– Regular expressions – File operations.
10	**Unit-X: WEB TECHNOLOGIES** Internet fundamentals – HTML, Tags with Attributes: <h1> to <h6>, <q>, <strong>, <big>, <small>, <ins>, <del>, <b>, <i>, <u>, <strike>, <sub>, <sup>,<font>, <marquee>, <table>, <form>, <frame> tags – Cascading Style Sheets (CSS) –XML – Java script: data types, operators, conditional and iteration statements, procedures, functions and arrays – PHP: data types, variables, operators, conditional and looping statements, strings and string methods, arrays and Array class methods, functions, classes and objects – Concept of accessing databases – Sessions and Cookies.